

Appendix

Anomalies-by-Synthesis: Anomaly Detection using Generative Diffusion Models for Off-Road Navigation

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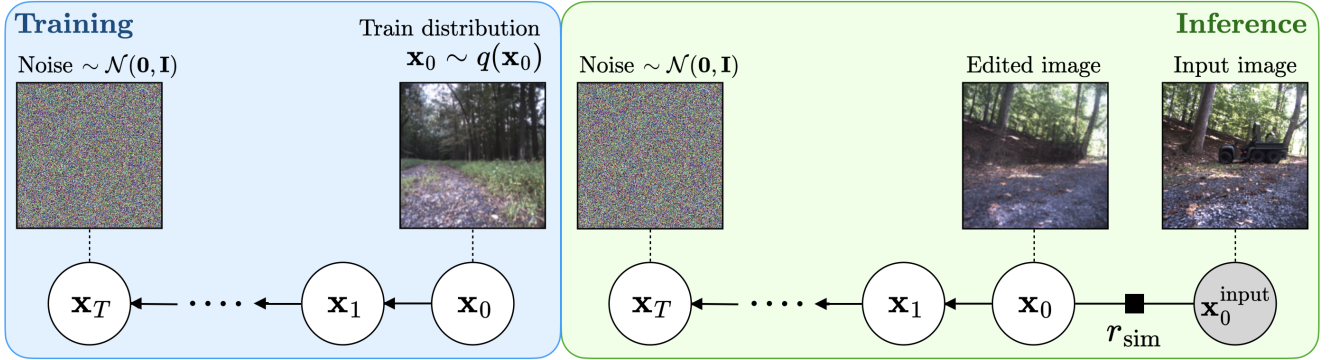


Fig. 4: Probabilistic graphical model of the conditional forward diffusion process. We wish to sample the random variable $\mathbf{x}_0$ corresponding to the training data distribution $q(\mathbf{x}_0)$. The *directed* edges between $\mathbf{x}_{t-1}$ and $\mathbf{x}_t$ (for $t = 1, \dots, T$) correspond to the vanilla forward diffusion process. Each directed edge denotes the sampling distribution $q(\mathbf{x}_t | \mathbf{x}_{t-1})$ which successively adds a small amounts of Gaussian noise: $q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$ [32, 38]. However, we are interested in sampling from $q(\mathbf{x}_0 | \mathbf{x}_0^{\text{input}}) = q(\mathbf{x}_0, r_{\text{sim}}(\mathbf{x}_0, \mathbf{x}_0^{\text{input}}))$. This objective corresponds to adding an additional *undirected factor* $r_{\text{sim}}(\mathbf{x}_0, \mathbf{x}_0^{\text{input}})$ between $\mathbf{x}_0$ and $\mathbf{x}_0^{\text{input}}$; $\mathbf{x}_0^{\text{input}}$ is treated as constant. Our task is to perform inference over this graphical model and sample from $q(\mathbf{x}_0 | \mathbf{x}_0^{\text{input}})$ using a diffusion model that was trained to perform reverse diffusion in the absence of the r_{sim} factor.

APPENDIX I

PROOF: GENERALIZED CONDITIONAL GUIDANCE GRADIENT

Classifier guidance is not only restricted to classifiers, it also requires training a classifier $p(y | \mathbf{x}_t)$ for each intermediate latent state [34, 39]. First, we extend classifier guidance to the more general setting where the conditioner is any non-negative function:

Theorem 1: When a diffusion model ϵ_t^θ is trained to sample from $q(\mathbf{x}_0)$, the conditional distribution $q(\mathbf{x}_0 | \mathbf{x}_0^{\text{input}}) \propto q(\mathbf{x}_0) r_{\text{sim}}(\mathbf{x}_0, \mathbf{x}_0^{\text{input}})$ can be sampled by using the following guidance gradient during reverse diffusion:

$$\begin{aligned} \mathbf{g}_t(\mathbf{x}_t) &= \nabla_{\mathbf{x}_t} \log \int_{\mathbf{x}_0} q(\mathbf{x}_0 | \mathbf{x}_t) r_{\text{sim}}(\mathbf{x}_0, \mathbf{x}_0^{\text{input}}) d\mathbf{x}_0 \\ &= \nabla_{\mathbf{x}_t} \log \mathbb{E}_{q(\mathbf{x}_0 | \mathbf{x}_t)} [r_{\text{sim}}(\mathbf{x}_0, \mathbf{x}_0^{\text{input}})] \end{aligned} \quad (4)$$

A. Proof using the variational inference perspective

$$q(\mathbf{x}_t | \mathbf{x}_{t+1}, \mathbf{x}_0^{\text{input}}) \propto q(\mathbf{x}_t, \mathbf{x}_{t+1} | \mathbf{x}_0^{\text{input}})$$

$$\begin{aligned} &= \int_{\mathbf{x}_0} q(\mathbf{x}_t, \mathbf{x}_{t+1}, \mathbf{x}_0 | \mathbf{x}_0^{\text{input}}) d\mathbf{x}_0 \\ &= \int_{\mathbf{x}_0} q(\mathbf{x}_0 | \mathbf{x}_0^{\text{input}}) q(\mathbf{x}_t, \mathbf{x}_{t+1} | \mathbf{x}_0, \mathbf{x}_0^{\text{input}}) d\mathbf{x}_0 \\ &= \int_{\mathbf{x}_0} q(\mathbf{x}_0 | \mathbf{x}_0^{\text{input}}) \underbrace{q(\mathbf{x}_t, \mathbf{x}_{t+1} | \mathbf{x}_0)}_{\text{(since } \mathbf{x}_{1:T} \text{ is independent of } \mathbf{x}_0^{\text{input}} \text{ conditioned on } \mathbf{x}_0)} d\mathbf{x}_0 \\ &\propto \int_{\mathbf{x}_0} \underbrace{q(\mathbf{x}_0) r_{\text{sim}}(\mathbf{x}_0, \mathbf{x}_0^{\text{input}})}_{\text{(by the definition of } q(\mathbf{x}_0 | \mathbf{x}_0^{\text{input}}) \propto q(\mathbf{x}_0) r_{\text{sim}}(\mathbf{x}_0, \mathbf{x}_0^{\text{input}}))} q(\mathbf{x}_t, \mathbf{x}_{t+1} | \mathbf{x}_0) d\mathbf{x}_0 \\ &= \int_{\mathbf{x}_0} r_{\text{sim}}(\mathbf{x}_0, \mathbf{x}_0^{\text{input}}) q(\mathbf{x}_t, \mathbf{x}_{t+1}, \mathbf{x}_0) d\mathbf{x}_0 \\ &= \int_{\mathbf{x}_0} r_{\text{sim}}(\mathbf{x}_0, \mathbf{x}_0^{\text{input}}) q(\mathbf{x}_{t+1}) q(\mathbf{x}_t | \mathbf{x}_{t+1}) q(\mathbf{x}_0 | \mathbf{x}_t, \mathbf{x}_{t+1}) d\mathbf{x}_0 \\ &= \int_{\mathbf{x}_0} r_{\text{sim}}(\mathbf{x}_0, \mathbf{x}_0^{\text{input}}) q(\mathbf{x}_{t+1}) q(\mathbf{x}_t | \mathbf{x}_{t+1}) \underbrace{q(\mathbf{x}_0 | \mathbf{x}_t)}_{\text{(since } \mathbf{x}_0 \text{ and } \mathbf{x}_{t+1} \text{ are independent conditioned on } \mathbf{x}_t)} d\mathbf{x}_0 \\ &\propto q(\mathbf{x}_t | \mathbf{x}_{t+1}) \int_{\mathbf{x}_0} q(\mathbf{x}_0 | \mathbf{x}_t) r_{\text{sim}}(\mathbf{x}_0, \mathbf{x}_0^{\text{input}}) d\mathbf{x}_0 \\ &= q(\mathbf{x}_t | \mathbf{x}_{t+1}) \mathbb{E}_{q(\mathbf{x}_0 | \mathbf{x}_t)} [r_{\text{sim}}(\mathbf{x}_0, \mathbf{x}_0^{\text{input}})] \end{aligned}$$

Therefore, following the same analysis as the first-order Gaussian approximation, the guidance gradient is $\nabla_{\mathbf{x}_t} \log \mathbb{E}_{q(\mathbf{x}_0 | \mathbf{x}_t)} [r_{\text{sim}}(\mathbf{x}_0, \mathbf{x}_0^{\text{input}})]$ evaluated at $\mathbf{x}_t = \mu_{t+1}(\mathbf{x}_{t+1})$.

B. Proof using the score functions perspective

The score function of intermediate states $\mathbf{x}_t$ of the vanilla forward diffusion process is defined as $\mathbf{s}(\mathbf{x}_t) = \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t)$. However, we're interested in the score function of the *conditional* forward diffusion process $\mathbf{s}(\mathbf{x}_t | \mathbf{x}_0^{\text{input}}) = \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0^{\text{input}})$. The additional term that needs to be added to $\mathbf{s}(\mathbf{x}_t)$ to obtain $\mathbf{s}(\mathbf{x}_t | \mathbf{x}_0^{\text{input}})$ is the guidance gradient. Therefore, we now derive $\mathbf{s}(\mathbf{x}_t | \mathbf{x}_0^{\text{input}})$ in terms of $\mathbf{s}(\mathbf{x}_t)$:

$$\begin{aligned}
\underbrace{\mathbf{s}(\mathbf{x}_t | \mathbf{x}_0^{\text{input}})}_{\text{conditional score function}} &= \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0^{\text{input}}) \\
&= \nabla_{\mathbf{x}_t} \log \int_{\mathbf{x}_0} q(\mathbf{x}_t, \mathbf{x}_0 | \mathbf{x}_0^{\text{input}}) d\mathbf{x}_0 \\
&= \nabla_{\mathbf{x}_t} \log \int_{\mathbf{x}_0} q(\mathbf{x}_0 | \mathbf{x}_0^{\text{input}}) q(\mathbf{x}_t | \mathbf{x}_0, \mathbf{x}_0^{\text{input}}) d\mathbf{x}_0 \\
&= \nabla_{\mathbf{x}_t} \log \int_{\mathbf{x}_0} q(\mathbf{x}_0 | \mathbf{x}_0^{\text{input}}) \underbrace{q(\mathbf{x}_t | \mathbf{x}_0)}_{\text{(since } \mathbf{x}_{1:T} \text{ is independent of } \mathbf{x}_0^{\text{input}} \text{ conditioned on } \mathbf{x}_0)} d\mathbf{x}_0 \\
&= \nabla_{\mathbf{x}_t} \log \int_{\mathbf{x}_0} \underbrace{q(\mathbf{x}_0) r_{\text{sim}}(\mathbf{x}_0, \mathbf{x}_0^{\text{input}})}_{\text{(by the definition of } q(\mathbf{x}_0 | \mathbf{x}_0^{\text{input}}) \propto q(\mathbf{x}_0) r_{\text{sim}}(\mathbf{x}_0, \mathbf{x}_0^{\text{input}}))} q(\mathbf{x}_t | \mathbf{x}_0) d\mathbf{x}_0 \\
&= \nabla_{\mathbf{x}_t} \log \int_{\mathbf{x}_0} r_{\text{sim}}(\mathbf{x}_0, \mathbf{x}_0^{\text{input}}) q(\mathbf{x}_t, \mathbf{x}_0) d\mathbf{x}_0 \\
&= \nabla_{\mathbf{x}_t} \log \int_{\mathbf{x}_0} r_{\text{sim}}(\mathbf{x}_0, \mathbf{x}_0^{\text{input}}) q(\mathbf{x}_t) q(\mathbf{x}_0 | \mathbf{x}_t) d\mathbf{x}_0 \\
&= \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log \int_{\mathbf{x}_0} q(\mathbf{x}_0 | \mathbf{x}_t) r_{\text{sim}}(\mathbf{x}_0, \mathbf{x}_0^{\text{input}}) d\mathbf{x}_0 \\
&= \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log \mathbb{E}_{q(\mathbf{x}_0 | \mathbf{x}_t)} [r_{\text{sim}}(\mathbf{x}_0, \mathbf{x}_0^{\text{input}})] \\
&= \mathbf{s}(\mathbf{x}_t) + \underbrace{\nabla_{\mathbf{x}_t} \log \mathbb{E}_{q(\mathbf{x}_0 | \mathbf{x}_t)} [r_{\text{sim}}(\mathbf{x}_0, \mathbf{x}_0^{\text{input}})]}_{\text{guidance gradient}}
\end{aligned}$$

APPENDIX II

TRAINING ON THE RUGD DATASET

A. Information about the RUGD dataset

The RUGD dataset (Fig. 5, [15]) is an off-road dataset of video sequences captured from a small, unmanned mobile robot traversing in unstructured environments. It contains over 7,453 labeled images from 17 scenes, annotated with pixel-level segmentation over 24 semantic classes. The annotated frames are spaced five frames apart.

We split the 24 semantic categories as 16 in-distribution labels: $\mathbb{C}_{\text{ID}} = \{\text{dirt, sand, grass, tree, pole, sky, asphalt, gravel, mulch, rock-bed, log, fence, bush, sign, rock, concrete}\}$, and 8 OOD labels corresponding to ‘‘obstacle’’ classes: $\mathbb{C}_{\text{OOD}} = \{\text{vehicle, container/generic-object, building, bicycle, person, bridge, picnic-table, water}\}$

B. Training a diffusion model on the RUGD Dataset

Our diffusion model is trained on samples from the RUGD train split that does not contain humans and artificial constructs (Fig. 6 (left)). The out-of-distribution and anomalous images are ‘held out’ for evaluation (Fig. 6 (right)). As can be seen from Fig. 7, our trained diffusion model successfully generates realistic images containing only in-distribution classes such as trees, grass, ground *etc.*

C. Tuning the Guidance Strength

The strength of the guidance term in our diffusion model can be tuned to enforce a variable level of consistency between the input image and the image synthesized by the diffusion model. Fig. 8 shows the impact of the guidance term on the image generated, for an example input image from the RUGD dataset.

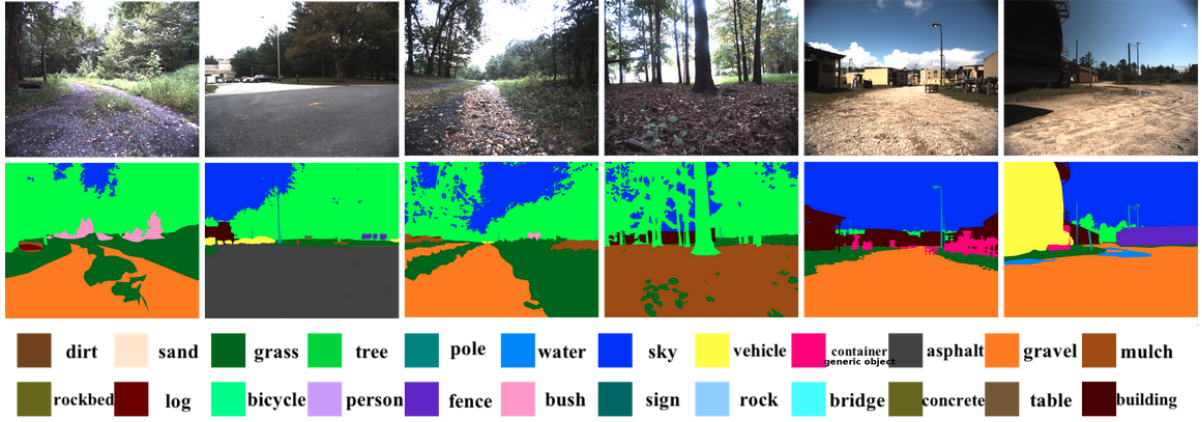


Fig. 5: Examples of video frames, annotations and semantic classes from the full RUGD dataset [15].



Fig. 6: In-distribution and out-of-distribution images from the RUGD dataset. *Left:* Examples of the in-distribution images on which our RUGD diffusion model was trained. In general, these images contain natural, off-road vegetation — a mixture of forest, meadow, mulch, and paths, without any humans or artificial constructions like buildings or vehicles. *Right:* Examples of held-out, out-of-distribution RUGD images the robot might encounter. These contains anomaly objects like buildings and vehicles. The diffusion model trained on the images on the left must remove anomalies from the images on the right.



Fig. 7: Samples generated from the trained diffusion model (without conditioning). The training data is shown in Fig. 6 (left). The generated samples are photorealistic, and appear very similar to the training images.

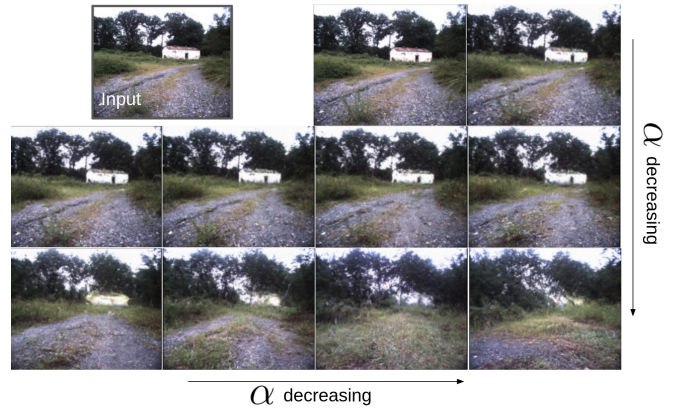


Fig. 8: The strength of the diffusion model's guidance gradient can be tuned by a hyperparameter α . As α decreases, the guidance enforcing similarity between the input image and the image synthesized by the model reduces.